Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

Spring 2026

Problem Set #4

Posted: Monday, Feb 23 **Due: Tuesday, Mar 3, 11:59 pm**

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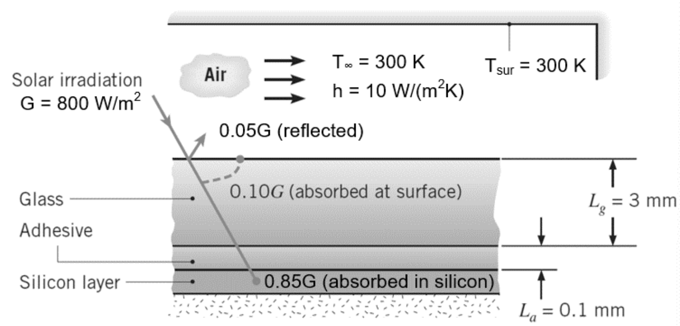
1. (40 pt) A photovoltaic (PV) panel consists of (top to bottom) a 3-mm-thick ceria-doped glass (k_g = 1.5 W/m·K), a 0.1-mm-thick optical grade adhesive (k_a = 1 W/m·K), and a very thin layer of silicon within which solar energy is converted to electricity. The solar-to-electricity conversion efficiency within the silicon layer η decreases with increasing silicon temperature T_{Si}, as described by the expression $\eta = a - bT_{Si}$, where a = 0.553 and b = 0.001 K^{-1}. The temperature T_{Si} is expressed in Kelvins. The backside of the silicon layer is thermally insulated. Contact thermal resistance between the adhesive and silicon layer is $R_c = 10^{-4}$ m^2·K/W. There is no contact resistance between the glass and the adhesive.



In a sunny day, there is G = 800 W/m^2 solar irradiation reaching the PV panel. 5% of solar irradiation is reflected from the top surface of the glass, 10% of solar irradiation is absorbed at the top surface of the glass, and 85% of solar irradiation is transmitted to and absorbed within the silicon layer. Part of the solar irradiation absorbed in the silicon is converted to thermal energy and the remainder is converted to electricity. The PV panel is exposed to ambient air with temperature of T_∞ = 300 K and convective heat transfer coefficient of h = 10 W/m^2·K. The surrounding temperature is $T_{sur} = T_\infty$ = 300 K and the effective radiative heat transfer coefficient is h_{rad} = 5 W/m^2·K. Hint: you do not need to use the Stefan-Boltzmann law to calculate the radiative heat loss.

(a) Show the thermal resistance network that can describe heat transfer across the PV panel. Clearly label all thermal resistances, nodes for temperature, and heat fluxes entering the temperature nodes. State key assumptions that make the thermal resistance network valid. Calculate the value of each thermal resistance in the network.

(b) Calculate the temperatures of the top surface of the glass $T_{g,top}$ and the silicon layer T_{Si}.

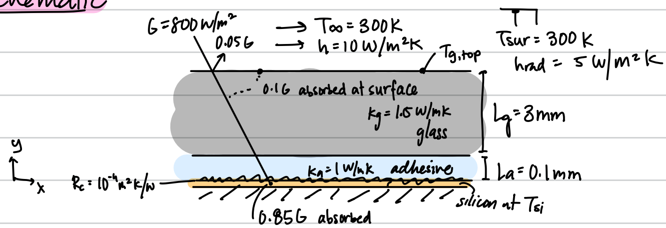
(c) Calculate the solar-to-electricity conversion efficiency η and resulting electricity power flux produced by the PV panel.

Problem #1

Known: $t_g = 3mm$, $t_a = 0.1\,mm$, $\eta = 0.553 - (0.001\,K^{-1})T_{si}$, $R_c = 10^{-4}\,m^2K/W$
$G = 800\,W/m^2$, $0.05G$ reflected, $0.1G$ absorbed at surface, $0.85G$ absorbed in silicon, $T_\infty = 300K = T_{sur}$

Find: a) thermal resistance network across PV panel
b) $T_{g,top}$, T_{si}
c) η and power flux produced by PV panel

Schematic:



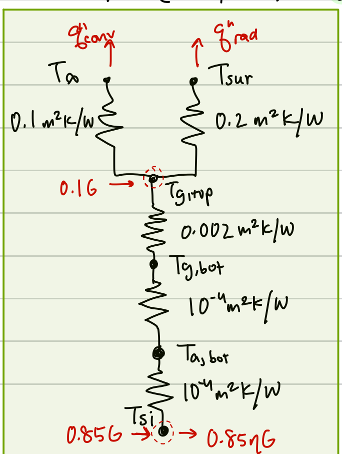
Properties: $k_g = 1.5\,W/mK$, $k_a = 1\,W/mK$, $h = 10\,W/m^2K$, $h_{rad} = 5\,W/m^2K$

Assumptions: negligible silicon thickness, perfectly thermally insulated, no contact resistance between glass & adhesive, uniform constant material properties & T_{sur}, no volumetric heat generation, 1D heat transfer, steady state

Analysis: a) There are 5 thermal resistances:

1) $R''_c = 10^{-4}\,m^2K/W$
2) $R''_{a,cond} = \frac{L_a}{k_a} = (0.1\times10^{-3}\,m)(1\,W/mK)^{-1} = 1\times10^{-4}\,m^2K/W$
3) $R''_{g,cond} = \frac{L_g}{k_g} = (3\times10^{-3}\,m)(1.5\,W/mK)^{-1} = 0.002\,m^2K/W$
4) $R''_{rad} = \frac{1}{h_{rad}} = (5\,W/m^2K)^{-1} = 0.2\,m^2K/W$
5) $R''_{conv} = \frac{1}{h} = (10\,W/m^2K)^{-1} = 0.1\,m^2K/W$

1)–3) in series; 4)–5) in parallel; in series



Key assumptions: · 1D heat transfer
· no volumetric heat generation
· steady state
· adiabatic base
· negligibly thin silicon

*Referenced example 3.3 in Fundamentals of Heat & Mass Transfer

b) Conservation of energy on node $T_{g,top}$ gives:

$$0.1G + q''_{g,cond} - q''_{conv} - q''_{rad} = 0$$

Let $R''_{int} = R''_c + R''_{a,cond} + R''_{g,cond} = (10^{-4} + 10^{-4} + 0.002)\ m^2K/W = 0.0022\ m^2K/W$

$$R''_{env} = \left(\frac{1}{R''_{conv}} + \frac{1}{R''_{rad}}\right)^{-1} = \left(\frac{1}{0.2} + \frac{1}{0.1}\ W/m^2K\right)^{-1} = \frac{1}{15}\ m^2K/W$$

Since $q'' = \frac{\Delta T}{R''}$ (and $T_{surr} = T_\infty$):

$$0.1G + \frac{T_{si} - T_{g,top}}{R''_{int}} - \frac{T_{gtop} - T_\infty}{R''_{env}} = 0$$

$$0.1G + \frac{T_{si}}{R''_{int}} - T_{gtop}\left(\frac{1}{R''_{int}} + \frac{1}{R''_{env}}\right) + \frac{T_\infty}{R''_{env}} = 0$$

Conservation of energy on node T_{si} gives:

$$0.85G - 0.85\eta G - \frac{T_{si} - T_{g,top}}{R''_{int}} = 0.85 - 0.85(a - bT_{si})G - \frac{T_{si} - T_{gtop}}{R''_{int}} = 0$$

Solving for T_{si} in terms of T_{gtop}:

$$\left[0.85G - 0.85aG + 0.85bG\,T_{si} - \frac{T_{si}}{R''_{int}} + \frac{T_{gtop}}{R''_{int}} = 0\right] R''_{int}$$

$$0.85GR''_{int}(1-a) + 0.85bGR''_{int}T_{si} - T_{si} + T_{gtop} = 0$$

$$T_{si} = \frac{T_{gtop} + 0.85GR''_{int}(1-a)}{1 - 0.85bGR''_{int}}$$

plug into first energy balance:

$$0.1G - T_{gtop}\left(\frac{1}{R''_{int}} + \frac{1}{R''_{env}}\right) + \frac{1}{R''_{int}}\left(\frac{T_{gtop} + 0.85GR''_{int}(1-a)}{1 - 0.85bGR''_{int}}\right) + \frac{T_\infty}{R''_{env}} = 0$$

$$0.1G + \left(0.85G(1-a)(1 - 0.85bGR''_{int})^{-1}\right) + \frac{T_\infty}{R''_{env}} = T_{gtop}\left(\frac{1}{R''_{int}} + \frac{1}{R''_{env}}\right) - T_{gtop}\left(R''_{int} - 0.85bGR''^{2}_{int}\right)^{-1}$$

$$T_{gtop} = \frac{0.1G + 0.85G(1-a)(1 - 0.85bGR''_{int})^{-1} + \frac{T_\infty}{R''_{env}}}{\left(\frac{1}{R''_{int}} + \frac{1}{R''_{env}}\right) - \left(R''_{int} - 0.85bGR''^{2}_{int}\right)^{-1}}$$

$$T_{gtop} = \frac{0.1(800\ W/m^2) + 0.85(800\ W/m^2)(1 - 0.553)(1 - 0.85\cdot 0.001\ K^{-1}\cdot 800\ W/m^2\cdot 0.0022\ m^2K/W)^{-1} + 300K(15\ W/m^2K)}{\left(\frac{1}{0.0022} + 15\right)W/m^2K - \left(0.0022\ m^2K/W - 0.85(0.001\ K^{-1})(800\ W/m^2)(0.0022\ m^2K)^2\right)^{-1}}$$

$T_{gtop} = 341.115\ K$

$$T_{si} = \frac{341.115\ K + 0.85(800\ W/m^2)(0.0022\ m^2K/W)(1 - 0.553)}{1 - 0.85(0.001K^{-1})(800\ W/m^2)(0.0022\ m^2K/W)}$$

$T_{si} = 342.296\ K$

c) Since $T_{si} = 342.296\ K$:

$$\eta = a - b\,T_{si} = 0.553 - (0.001\ K^{-1} \cdot 342.296\ K)$$
$$= 0.211 = 21.1\%$$

Power flux:

$$P'' = 0.85\,\eta\,G = 0.85(0.211)(800\ W/m^2)$$
$$= 143.48\ W/m^2$$